

If the investor's trade schedule sequence followed a different weighting scheme than volume weighting, such as front-or back-loaded weightings, it would be prudent for investors to compute the index cost in each period. In times where the index VWAP is not available, it can be approximated as $\text{Index VWAP} = 1/2 \cdot R_m$, where R_m is the total return in basis points of the underlying index over the period. The $1/2$ is the adjustment factor to account for continuous trading Kissell (2008).

The index adjusted cost is then:

$$\text{Index Adjusted Cost}_{bp} = \text{Arrival Cost}_{bp} - \hat{b}_{KI} \cdot \text{Index Cost}_{bp} \quad (3.22)$$

$\hat{b}_{KI}$ is the stock k's sensitivity to the index. It is determined via linear regression in the same manner we calculate beta to the market index. Notice all we have done is subtract out the movement in the stock price that we would have expected to occur based only on the index movement. The index cost is not adjusted for the side of the trade.

Z-Score Evaluation Metric

The z-score evaluation metric provides a risk adjusted performance score by normalizing the difference between estimated and actual by the timing risk of the execution. This provides a normalized score that can be compared across different stocks and across days. (A z-score measure is also used to measure the accuracy of pre-trade models and to determine if these models are providing reasonable insight to potential outcomes cost.)

A simple statistical z-score is calculated as follows:

$$Z = \frac{\text{Actual} - \text{Expected}}{\text{Standard Deviation}}$$

For transaction cost analysis, we compute the normalized transaction cost as follows:

$$Z = \frac{\text{Pre-Trade Cost Estimate} - \text{Arrival Cost}}{\text{Pre-Trade Timing Risk}} \quad (3.23)$$

Notice that this representation is opposite the statistical z-score measure ($z = (x-u)/\sigma$). In our representation a positive z-score implies performance better than the estimate and a negative value implies performance worse than the estimate. Dividing by the timing risk of the trade normalizes for overall uncertainty due to price volatility and liquidity risk. This ensures that the sign of our performance metrics are consistent—positive indicates better performance and negative indicates lower quality performance.